

Hashing Variants and Addressing Techniques

Hashing techniques can be broadly classified into two categories based on how they handle collisions: **open addressing** and **closed addressing**. These relate closely to the concepts of **open hashing** and **closed hashing**.

Closed Addressing (Chaining)

Closed addressing stores all records that hash to the same location in a linked structure.

- Each bucket contains a linked list of keys with the same hash value.
- Easy insertion and deletion.
- Efficient when collisions are moderate.

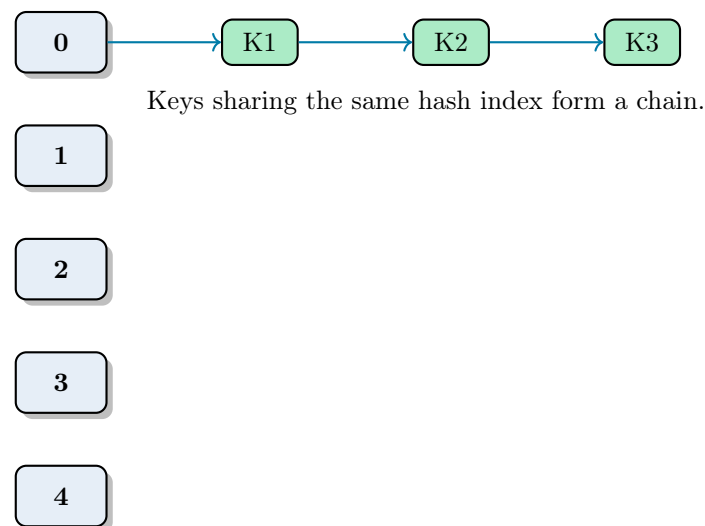


Figure 1: Closed Addressing: Chaining keys in linked lists.

Open Addressing

All records are stored directly in the table. Collisions are resolved by probing.

- No linked lists.
- Probe next slot if collision occurs.
- Deletion requires special handling.

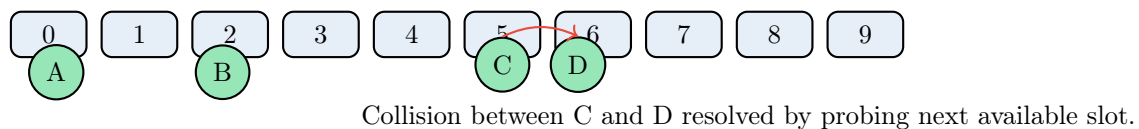


Figure 2: Open Addressing: Linear probing resolves collisions.

Linear Probing

Linear Probing Function

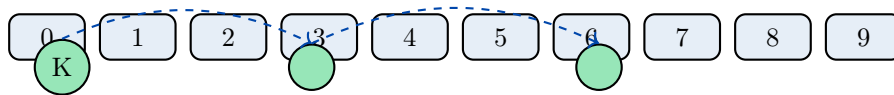
$$h_i(k) = (h(k) + i) \bmod m$$

Sequential probing can cause primary clustering.

Double Hashing

Double Hashing Function

$$h_i(k) = (h_1(k) + i \cdot h_2(k)) \bmod m$$



Each probe step is determined by $h_2(k)$, reducing clustering.

Figure 3: Double Hashing: Secondary hash function determines probe steps.

Comparison Summary

Technique	Collision Handling	Common Use
Closed Addressing	Chaining (linked list)	Database indexing, disk-based systems
Open Addressing	Probe next slot	In-memory hash tables
Linear Probing	Sequential slot checking	Simple, prone to clustering
Quadratic Probing	Quadratic offset probing	Reduces primary clustering
Double Hashing	Secondary hash function	Better distribution, efficient lookup

Table 1: Summary of Hashing Techniques and Collision Handling